

CS & IT ENGINEERING



Computer Network

Application Layer

Lecture No. - 01



By - Abhishek Sir



Recap of Previous Lecture



Topic

TCP Options



Topics to be Covered



Topic

DNS



ABOUT ME



Hello, I'm **Abhishek**

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- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : Application Layer



Application Layer Protocols :

- DNS : Domain Name System
- HTTP : Hyper Text Transfer Protocol
- FTP : File Transfer Protocol
- SMTP : Simple Mail Transfer Protocol

client-server
Model



Topic : DNS



→ Domain Name System (DNS)

→ 'Stateless' Application Layer Protocol

* DNS server is not maintains any state regarding DNS client

* Every DNS query is treated as new by DNS server.



Topic : DNS



→ Used to translate 'Host Name' into 'IP Address' (vice versa)

→ Hosts (DNS clients) and DNS server communicate to resolve names
[Name / Address translation] → Name Resolution



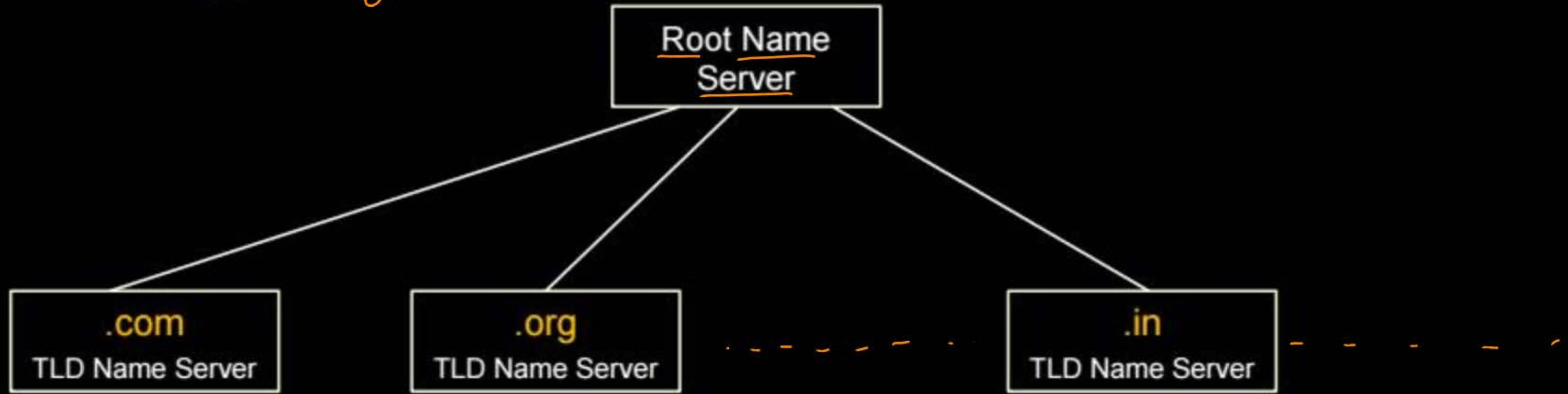
Topic : DNS Hierarchy



- Distributed Hierarchical Database : [Global Hierarchy]
[Implemented in hierarchy of many name servers]



Topic : DNS Hierarchy





Topic : DNS Hierarchy



→ DNS name servers : 3-tier Hierarchy

1. Root Name Server 🏠
2. Top-Level Domain (TLD) Name Servers
3. Authoritative Name Servers



Topic : Root Name Servers



- Point of contact for any DNS query ✓
- ICANN manages root DNS domain
- 13 logical root name servers world wide
[each server replicated many times]

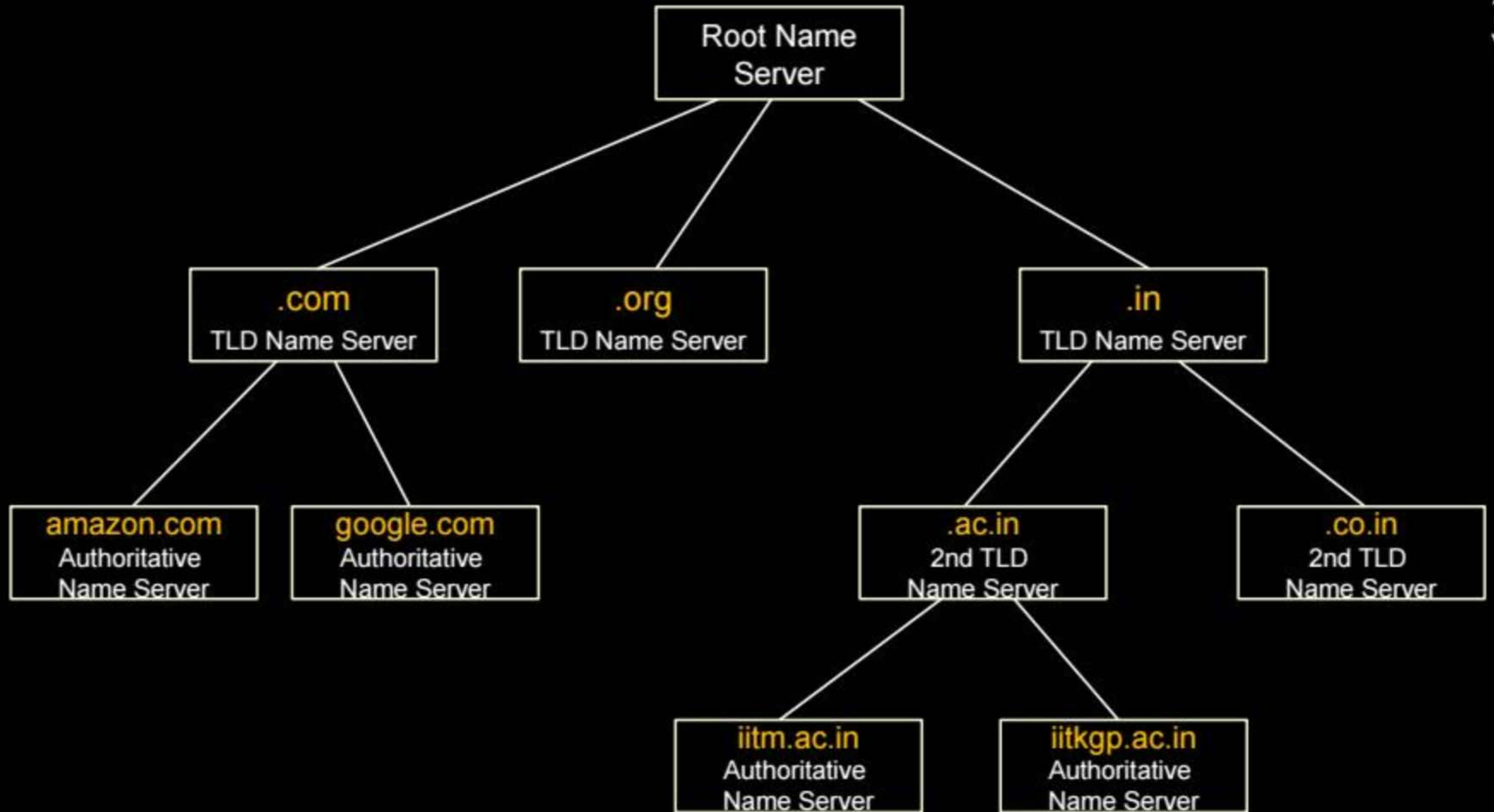


Topic : TLD Name Servers



→ Responsible for all top level domains like “.com”, “.org” etc
[Country domains e.g. “.in”, “.uk”, etc]

* 2nd TLD Name Server
→ Intermediate TLD
→ Sub-Domain





Topic : Authoritative Name Servers

- Organization's own DNS Server(s) ←
- Providing authoritative host name to IP mappings
- Can be maintained by organization or service provider



Topic : DNS Name Resolution

→ DNS Name Resolution or Lookup : 'DNS Resolver Program'

→ Two mode :-

1. Iterated Query *
2. Recursive Query ←



Topic : Iterated Query



→ Contacted server replies with name (address) of server to contact

Example :-

Domain Name : "www.amazon.com"

(DNS query)

[DNS Hierarchy]

(DNS client)

DNS
Resolver

Root Name
Server

.com
TLD Name Server

amazon.com
Authoritative
Name Server

- ① DNS query
- ② DNS Reply
- ③ DNS query
- ④ DNS Reply
- ⑤ DNS query
- ⑥ DNS Reply

Example 1:-

#Q. Consider the resolution of the domain name "www.amazon.com" by a DNS resolver. Assume that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is _____.

Ans = 3

Example 2:-

#Q. Consider the resolution of the domain name "www.amazon.com" by a DNS resolver. Assume that iterative query mechanism is used in the resolution. Let RTT denote the round trip time between (DNS resolver) and (DNS servers) in the DNS hierarchy. The number of RTTs required in completely resolving the domain name is _____.

$$\text{Ans} = [RTT_1 + RTT_2 + RTT_3] = 3RTT$$

$$RTT_1 = RTT_2 = RTT_3 = RTT$$

$$\boxed{\text{Ans} = 3}$$

[NAT]

(H.W.) IIT-KGP

[GATE-2022][1 Mark]



#Q. Consider the resolution of the domain name `www.gate.org.in` by a DNS resolver. Assume that no resource records are cached anywhere across the DNS servers and that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is _____.



Topic : Recursive Query



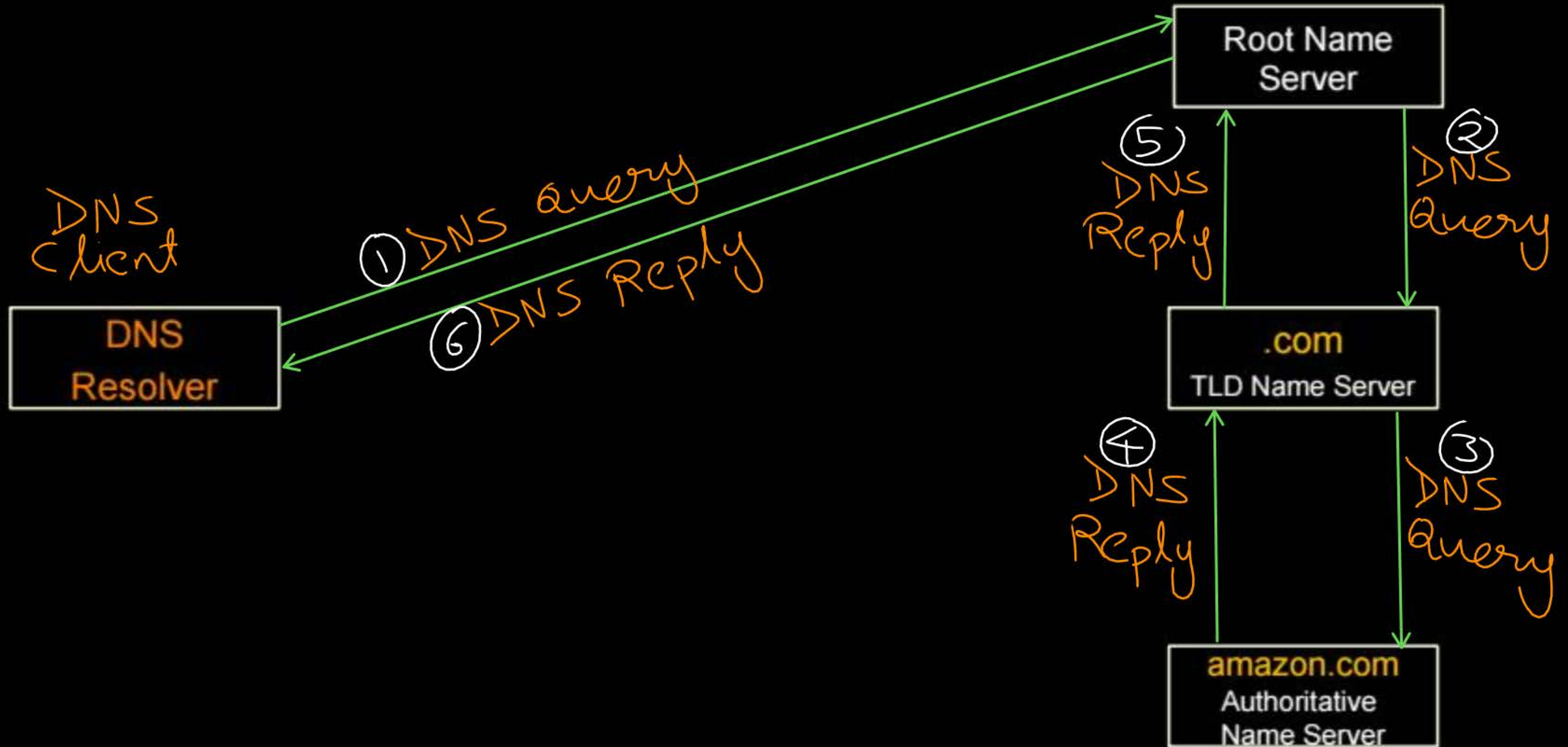
- Put burden of **name resolution** on **Contacted name server**
- Heavy load at **upper levels of hierarchy**

Example :-

Domain Name : "www.amazon.com"

DNS query

DNS Hierarchy





Topic : Local DNS Name Servers

→ Each ISP has local DNS name server

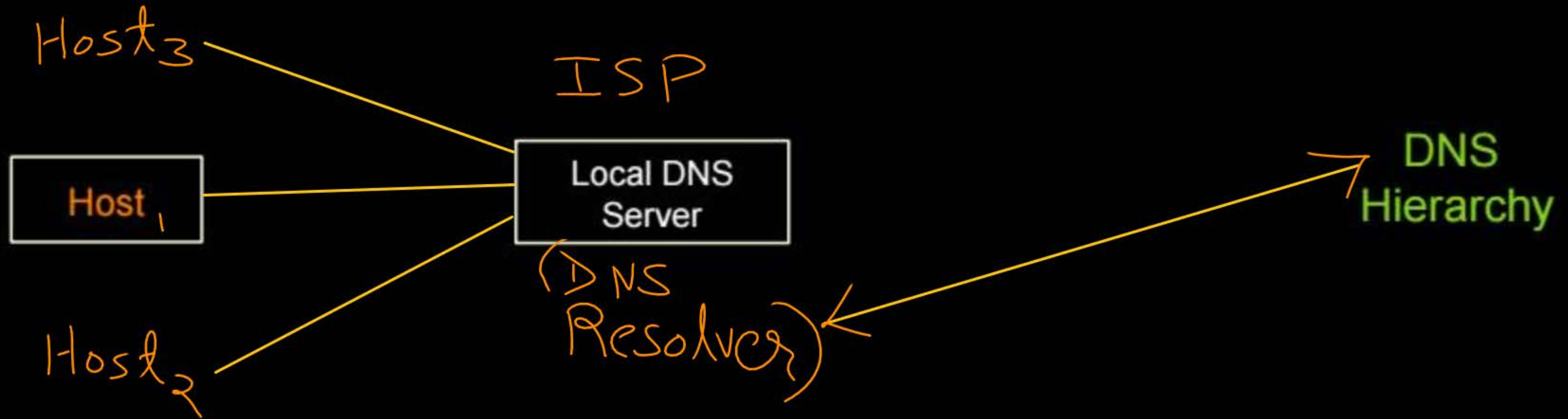
→ Local DNS server does not belongs to DNS hierarchy

→ subset of DNS hierarchy



Topic : Local DNS Name Servers

- When host make DNS query, it is sent to its local DNS server
- Local DNS server returns reply
- if Local DNS server does not have valid name-address pair
then it forward request into DNS hierarchy for resolution $\Rightarrow$ DNS Resolver





2 mins Summary



Topic

DNS



THANK - YOU

